RAMAYAN-GPT

RamayanaGPT delivers accurate, unbiased answers directly rooted in Valmiki's Ramayana, enhancing knowledge with authentic insights unavailable on conventional platforms.

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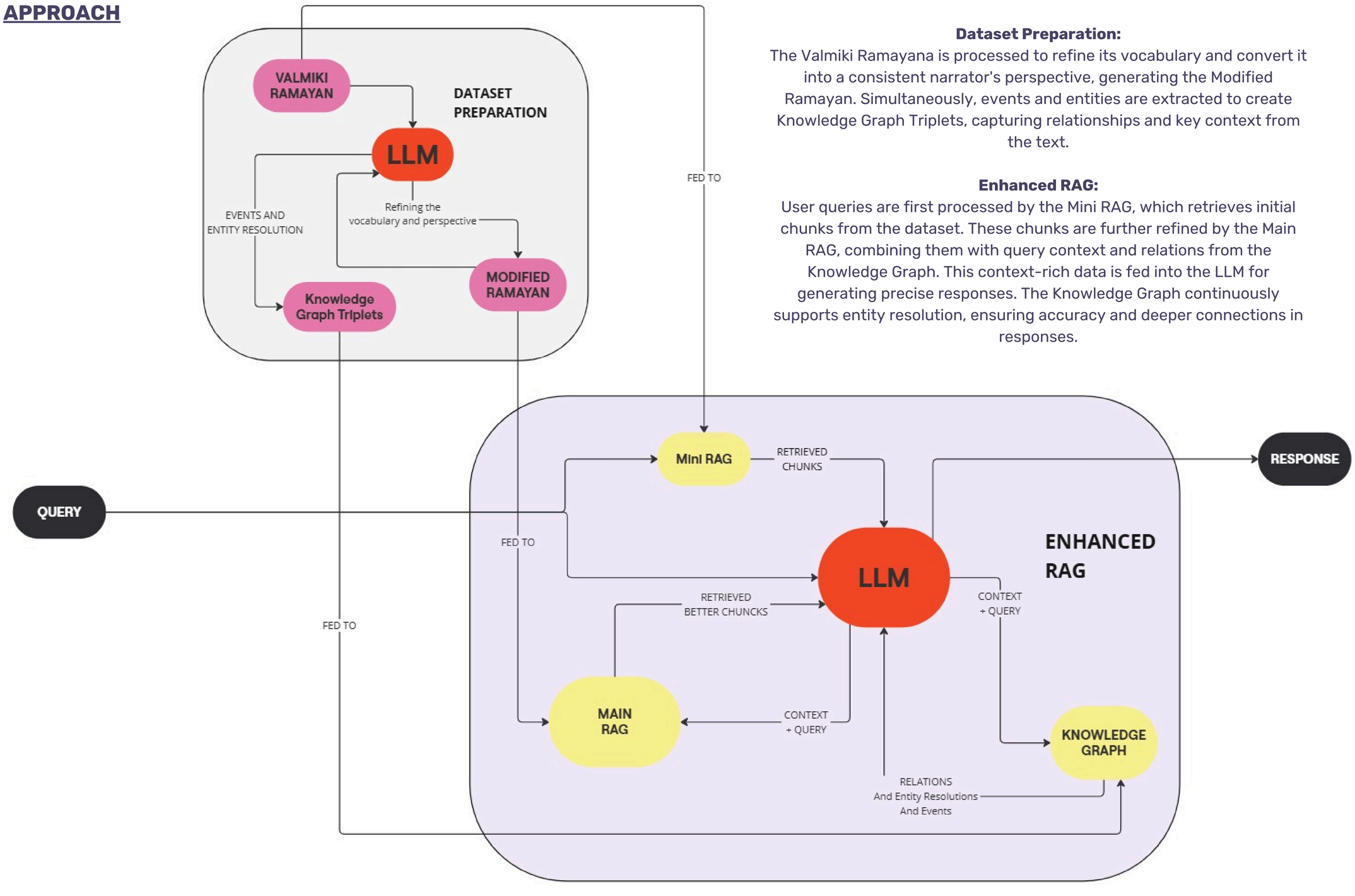
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Introduction

RamayanaGPT leverages AI to create an advanced Q&A system based on the Valmiki Ramayana. Using the original **Sanskrit and English** texts, it preprocesses the content to generate a modernized version in a narrator's perspective, optimized for large language models (LLMs). The project integrates **Retrieval-Augmented Generation** (RAG) and a **Knowledge Graph**, ensuring precise, context-aware answers. By blending ancient scripture with modern AI, RamayanaGPT preserves the authenticity of the Ramayana while enabling deeper exploration and understanding of this epic.



APPROACH



Dataset Preparation:
The Valmiki Ramayana is processed to refine its vocabulary and convert it into a consistent narrator's perspective, generating the Modified Ramayan. Simultaneously, events and entities are extracted to create Knowledge Graph Triplets, capturing relationships and key context from the text.

Enhanced RAG:
User queries are first processed by the Mini RAG, which retrieves initial chunks from the dataset. These chunks are further refined by the Main RAG, combining them with query context and relations from the Knowledge Graph. This context-rich data is fed into the LLM for generating precise responses. The Knowledge Graph continuously supports entity resolution, ensuring accuracy and deeper connections in responses.



CONCLUSION

RamayanaGPT brings the wisdom of the Valmiki Ramayana into modern AI, offering accurate and unbiased answers. By refining the text and using advanced retrieval methods, it preserves the original essence while making it accessible for today's audiences. This project highlights how AI can help preserve cultural heritage and enhance understanding of ancient epics.

FrameWorks

LLAMA-INDEX - RAG framework
GEMINI 1.5 - LLM
OCR FRAMEWORKS
Google Language Translator
Neo4j - KnowledgeGraph Database
Streamlit - Frontend
FastApi - For Api calls